

MAE 5730 HW Problem 12 (Ruina)

Eric Chiang

10/11/17

etc59

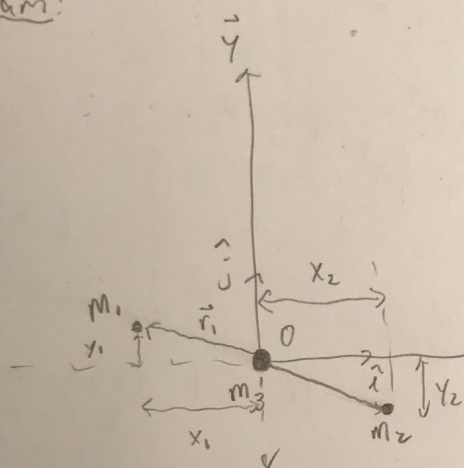
due: 10/14/17

I had my solution approved (first draft) by Professor Ruina on 10/12 office hours

12. Montgomery's Eight

Assume: Problem is in two dimensions
Only relevant force is inverse square gravity

Diagram:



Given (initial conditions):

$$t=0 (x_1, y_1) = (-0.97000436, 0.24308753)$$

$$t=0 (x_2, y_2) = (-x_1, -y_1)$$

$$t=0 (x_3, y_3) = (0, 0)$$

$$t=0 (v_{x3}, v_{y3}) = (0.93240737, 0.86473146)$$

$$(v_{x1}, v_{y1}) = (-v_{x3}/2, -v_{y3}/2)$$

$$(v_{x2}, v_{y2}) = (-v_{x3}/2, -v_{y3}/2)$$

Kinematics: all kinematics are relative to the origin of the fixed frame.

$$\vec{r}_1 = x_1 \hat{i} + y_1 \hat{j}$$

$$\dot{\vec{r}}_1 = \vec{v}_1 = \dot{x}_1 \hat{i} + \dot{y}_1 \hat{j}$$

$$\ddot{\vec{r}}_1 = \vec{a}_1 = \ddot{x}_1 \hat{i} + \ddot{y}_1 \hat{j}$$

$$\vec{r}_2 = x_2 \hat{i} + y_2 \hat{j}$$

$$\dot{\vec{r}}_2 = \vec{v}_2 = \dot{x}_2 \hat{i} + \dot{y}_2 \hat{j}$$

$$\ddot{\vec{r}}_2 = \vec{a}_2 = \ddot{x}_2 \hat{i} + \ddot{y}_2 \hat{j}$$

($\ddot{\vec{r}}_2$)

$$\vec{r}_3 = x_3 \hat{i} + y_3 \hat{j}$$

$$\dot{\vec{r}}_3 = \vec{v}_3 = \dot{x}_3 \hat{i} + \dot{y}_3 \hat{j}$$

$$\ddot{\vec{r}}_3 = \vec{a}_3 = \ddot{x}_3 \hat{i} + \ddot{y}_3 \hat{j} = (\ddot{\vec{r}}_3)$$

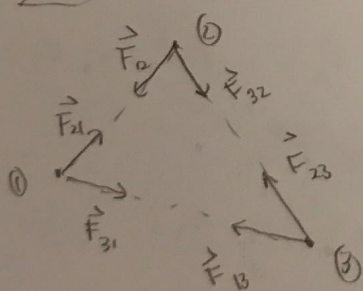
FBD:

Note: $\vec{F}_{31}$ = "force of mass 3 on mass 1"

By Newton III, $\vec{F}_{13} = -\vec{F}_{31}$
 $\vec{F}_{12} = -\vec{F}_{21}$

etc.

Note: FBD is in arbitrary (not to scale) configuration



gravitational

What direction are the forces in?

Force acts along line connecting the two masses. The direction dynamically changes as the masses move.

Let $\hat{e}_{12}$ be unit vector in direction point from mass 2 towards mass 1, etc

$$\hat{e}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|} = \frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = -\hat{e}_{21}$$

$$\hat{e}_{23} = \frac{\vec{r}_2 - \vec{r}_3}{|\vec{r}_2 - \vec{r}_3|} = -\hat{e}_{32}$$

$$\hat{e}_{13} = \frac{\vec{r}_{13}}{|\vec{r}_{13}|} = \frac{\vec{r}_1 - \vec{r}_3}{|\vec{r}_1 - \vec{r}_3|} = -\hat{e}_{31}$$

$$\vec{F}_{12} = \frac{Gm^2}{|\vec{r}_{12}|^2} \cdot \hat{e}_{12} = \frac{Gm^2}{|\vec{r}_2 - \vec{r}_1|^2} \cdot \frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{Gm^2}{|\vec{r}_2 - \vec{r}_1|^3} (\vec{r}_1 - \vec{r}_2) = -\vec{F}_{21}$$

similarly,

$$\vec{F}_{13} = \frac{Gm^2}{|\vec{r}_3 - \vec{r}_1|^3} (\vec{r}_1 - \vec{r}_3) = -\vec{F}_{31}$$

$$\vec{F}_{23} = \frac{Gm^2}{|\vec{r}_3 - \vec{r}_2|^3} (\vec{r}_2 - \vec{r}_3) = -\vec{F}_{32}$$

[note:]

$|\vec{r}_1 - \vec{r}_2| = |\vec{r}_2 - \vec{r}_1|$ (by geometry & vector properties)
etc

LMB: (mass 1)

$$\sum \vec{F}_{1, \text{tot}} = \vec{F}_{21} + \vec{F}_{31} = m \ddot{x}_1 \hat{i} + m \ddot{y}_1 \hat{j}$$

↳ sum of total forces on mass 1

$$= \frac{Gm^2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_2 - \vec{r}_1) + \frac{Gm^2}{|\vec{r}_1 - \vec{r}_3|^3} (\vec{r}_3 - \vec{r}_1) = m \ddot{\vec{r}}_1 \quad (1)$$

↳ force from m_2

↳ force from m_3

can eliminate since $m=1$
↳ solved for $\ddot{\vec{r}}_1$

$$\sum \vec{F}_{2, \text{tot}} = \vec{F}_{12} + \vec{F}_{32} = \frac{Gm^2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2) + \frac{Gm^2}{|\vec{r}_1 - \vec{r}_3|^3} (\vec{r}_3 - \vec{r}_2) = m_2 \ddot{\vec{r}}_2 \quad (2)$$

↳ force from m_1

↳ force from m_3

$$\sum \vec{F}_{3, \text{tot}} = \vec{F}_{13} + \vec{F}_{23} = \frac{Gm^2}{|\vec{r}_1 - \vec{r}_3|^3} (\vec{r}_1 - \vec{r}_3) + \frac{Gm^2}{|\vec{r}_2 - \vec{r}_3|^3} (\vec{r}_2 - \vec{r}_3) = m_3 \ddot{\vec{r}}_3 \quad (3)$$

Equations ①, ②, ③ represent six differential equations
(for variables $x_1, y_1, x_2, y_2, x_3, y_3$)

We can solve for the positions of the masses using MATLAB & ODE45

Note: We have simplified the DE's by analyzing the problem in the fixed frame and combining pairs of x & y variables into r .

$$\ddot{\mathbf{r}}_1 = \begin{bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{bmatrix} \quad \ddot{\mathbf{r}}_2 = \begin{bmatrix} \ddot{x}_2 \\ \ddot{y}_2 \end{bmatrix} \quad \ddot{\mathbf{r}}_3 = \begin{bmatrix} \ddot{x}_3 \\ \ddot{y}_3 \end{bmatrix}$$

$$(\vec{r}_2 - \vec{r}_1) = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix} \begin{bmatrix} \hat{i} & \hat{j} \end{bmatrix} \quad \dots \text{ similarly for any vector}$$

```

close all; clear all;
% MAE 5730 HW4 Problem 12
% Finding numerical solution Montgomery's eight problem
% Part a: run for 2.1 time units

tspan = [0, 2.1]; % initial and final times

% Establish parameters
p.m = 1;
p.G = 1;

% Initial Conditions
r0_1 = [-0.97000436, 0.24307653]';
r0_2 = -1.*r0_1;
r0_3 = [0 0]';

v0_3 = [0.93240737, 0.86473146]';
v0_1 = -0.5.*v0_3;
v0_2 = -0.5.*v0_3;

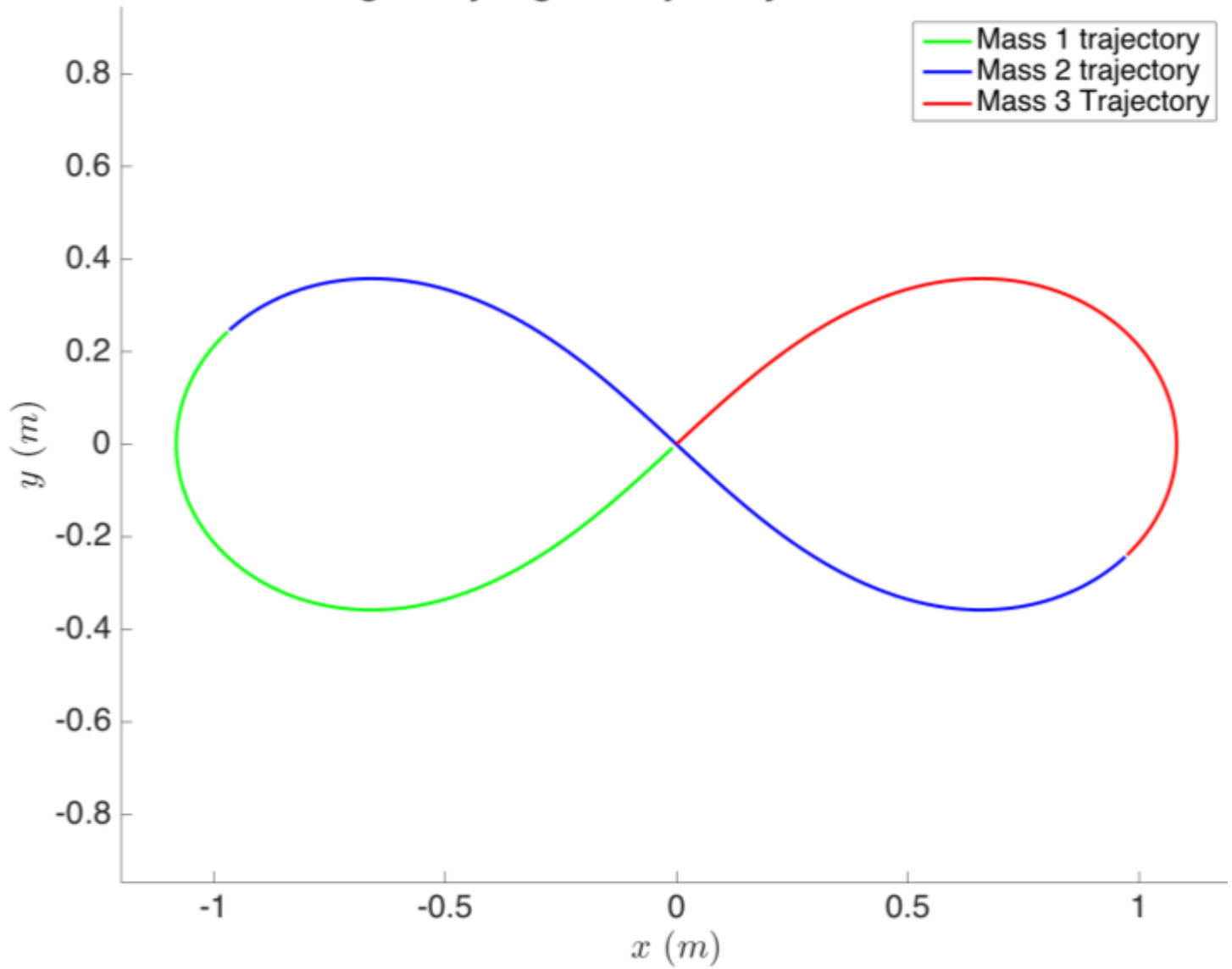
inits = [r0_1;r0_2;r0_3;v0_1;v0_2;v0_3];
opts.RelTol = 1e-6; opts.AbsTol = 1e-6;

% Integrate via ODE 45
[tarray,zarray] = ode45(@RHS_12,tspan,inits,opts,p);
x1_i = zarray(:,1); y1_i = zarray(:,2); % i stands for initial
x2_i = zarray(:,3); y2_i = zarray(:,4);
x3_i = zarray(:,5); y3_i = zarray(:,6);

% Plotting
figure
hold on
axis equal
xlim([-1.2 1.2])
r1 = plot(x1_i,y1_i,'g'); set(r1,{'LineWidth'},{2});
r2 = plot(x2_i,y2_i,'b'); set(r2,{'LineWidth'},{2});
r3 = plot(x3_i,y3_i,'r'); set(r3,{'LineWidth'},{2});
ylabel('$y$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
xlabel('$x$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
set(gca, 'fontsize', 20)
title('Montgomery Eight: Trajectory for 2.1 time units', 'FontSize', 20);
legend('Mass 1 trajectory', 'Mass 2 trajectory', 'Mass 3 Trajectory');

```

Montgomery Eight: Trajectory for 2.1 time units



p12a: We see that the three particles (labeled) move in such a way to form a figure 8.

```

% MAE 5730 HW4 Problem 12
% Finding numerical solution Montgomery's eight problem
% Part b: run for 10 time units

tspan = [0, 10]; % initial and final times

% Establish parameters
p.m = 1;
p.G = 1;

% Initial Conditions
r0_1 = [-0.97000436, 0.24307653]';
r0_2 = -1.*r0_1;
r0_3 = [0 0]';

v0_3 = [0.93240737, 0.86473146]';
v0_1 = -0.5.*v0_3;
v0_2 = -0.5.*v0_3;

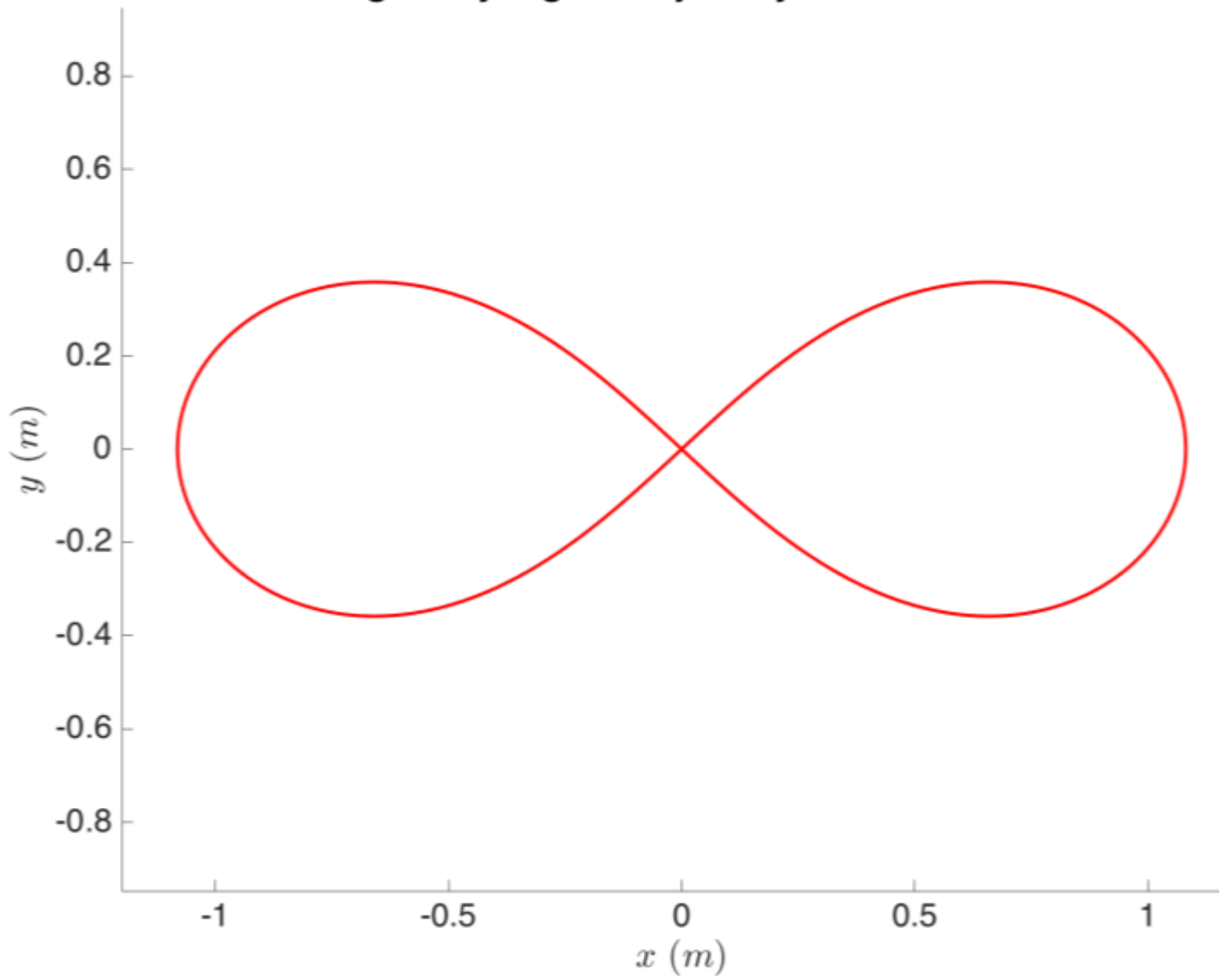
inits = [r0_1; r0_2; r0_3; v0_1; v0_2; v0_3];
opts.RelTol = 1e-6; opts.AbsTol = 1e-6;

% Integrate via ODE 45
[tarray, zarray] = ode45(@RHS_12, tspan, inits, opts, p);
x1 = zarray(:,1); y1 = zarray(:,2);
x2 = zarray(:,3); y2 = zarray(:,4);
x3 = zarray(:,5); y3 = zarray(:,6);

% Plotting
figure
hold on
axis equal
xlim([-1.2 1.2])
r1 = plot(x1, y1, 'g'); set(r1, {'LineWidth'}, {2});
r2 = plot(x2, y2, 'b'); set(r2, {'LineWidth'}, {2});
r3 = plot(x3, y3, 'r'); set(r3, {'LineWidth'}, {2});
ylabel('$y$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
xlabel('$x$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
set(gca, 'fontsize', 20)
title('Montgomery Eight: Trajectory for 10 time units', 'FontSize', 20);

```

Montgomery Eight: Trajectory for 10 time units



p12b: We see that after 10 time units, all the trajectories overlap with one another. This suggests that the figure 8 trajectory is a stable equilibrium trajectory for all three particles.

```

% MAE 5730 HW4 Problem 12
% Finding numerical solution Montgomery's eight problem
% Part c: run for 2.1 time units, slightly PERTURBED initial conditions

tspan = [0, 2.1]; % initial and final times

% Establish parameters
p.m = 1;
p.G = 1;

% Initial Conditions
r0_1 = [-0.97000436, 0.30]';
r0_2 = -1.*r0_1;
r0_3 = [0 0]';

v0_3 = [0.93240737, 0.87]';
v0_1 = -0.5.*v0_3;
v0_2 = -0.5.*v0_3;

inits_p = [r0_1; r0_2; r0_3; v0_1; v0_2; v0_3];
opts.RelTol = 1e-6; opts.AbsTol = 1e-6;

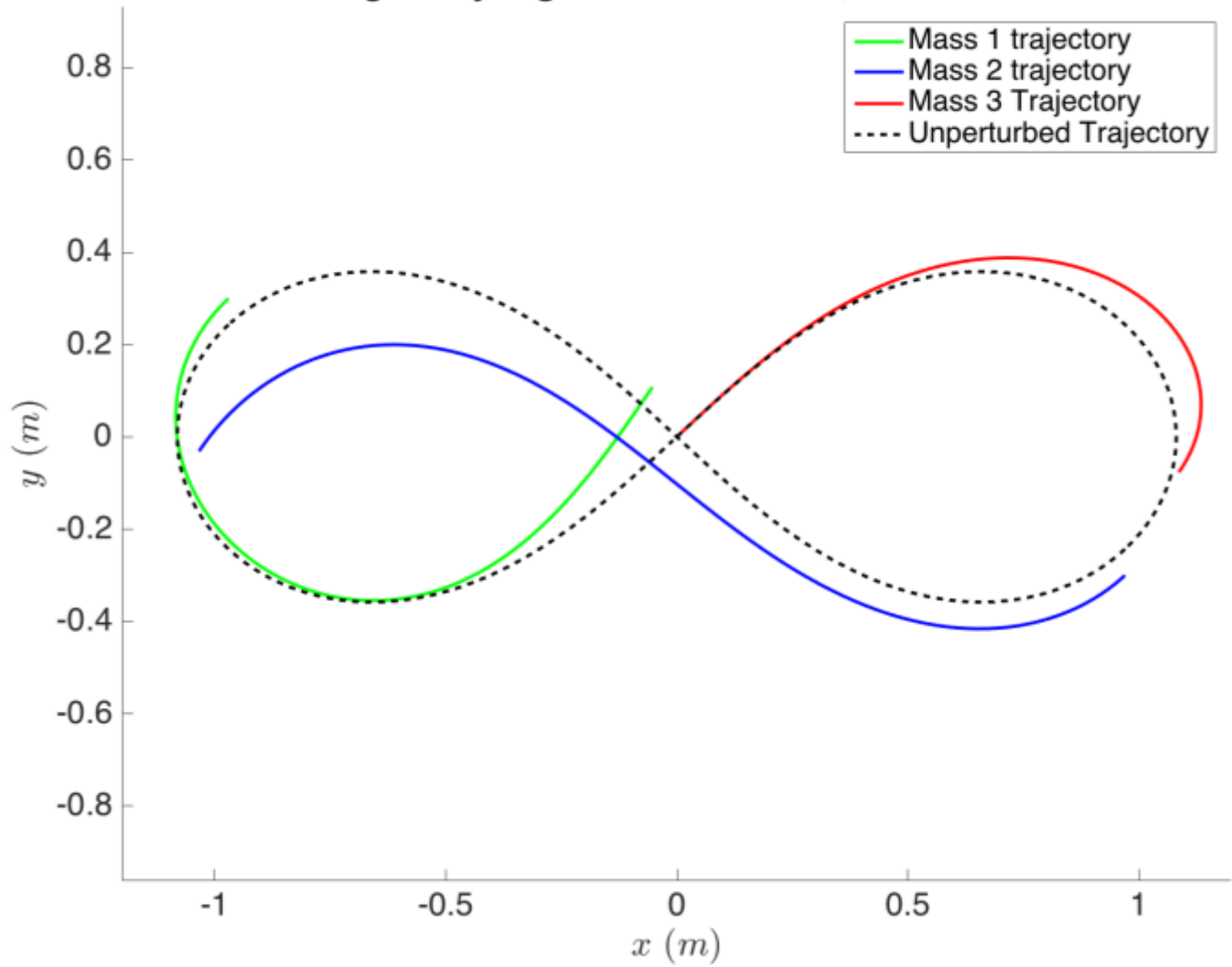
% Integrate via ODE 45
[tarray, zarray] = ode45(@RHS_12, tspan, inits_p, opts, p);
x1_p1 = zarray(:, 1); y1_p1 = zarray(:, 2);
x2_p1 = zarray(:, 3); y2_p1 = zarray(:, 4);
x3_p1 = zarray(:, 5); y3_p1 = zarray(:, 6);

% Plotting
figure
hold on
axis equal
xlim([-1.2 1.2])
r1 = plot(x1_p1, y1_p1, 'g'); set(r1, {'LineWidth'}, {2});
r2 = plot(x2_p1, y2_p1, 'b'); set(r2, {'LineWidth'}, {2});
r3 = plot(x3_p1, y3_p1, 'r'); set(r3, {'LineWidth'}, {2});
% For comparison
r1i = plot(x1_i, y1_i, 'k--'); set(r1i, {'LineWidth'}, {2});
r2i = plot(x2_i, y2_i, 'k--'); set(r2i, {'LineWidth'}, {2});
r3i = plot(x3_i, y3_i, 'k--'); set(r3i, {'LineWidth'}, {2});

ylabel('$y$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
xlabel('$x$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
set(gca, 'fontsize', 20)
title('Montgomery Eight: 2.1 time units, Perturbed IC', 'FontSize', 20);
legend('Mass 1 trajectory', 'Mass 2 trajectory', 'Mass 3 Trajectory', ...
      'Unperturbed Trajectory');

```


Montgomery Eight: 2.1 time units, Perturbed IC



p12c: We see that a small perturbation (as evidenced in the initial conditions), causes the trajectories to deviate from the original figure 8 shape we found in parts a. and b. of this problem.

```

% MAE 5730 HW4 Problem 12
% Finding numerical solution Montgomery's eight problem
% Part d: run for 10 time units, perturbed ICs

tspan = [0, 10]; % initial and final times

% Establish parameters
p.m = 1;
p.G = 1;

% Initial Conditions
r0_1 = [-0.97000436, 0.30]';
r0_2 = -1.*r0_1;
r0_3 = [0 0]';

v0_3 = [0.93240737, 0.87]';
v0_1 = -0.5.*v0_3;
v0_2 = -0.5.*v0_3;

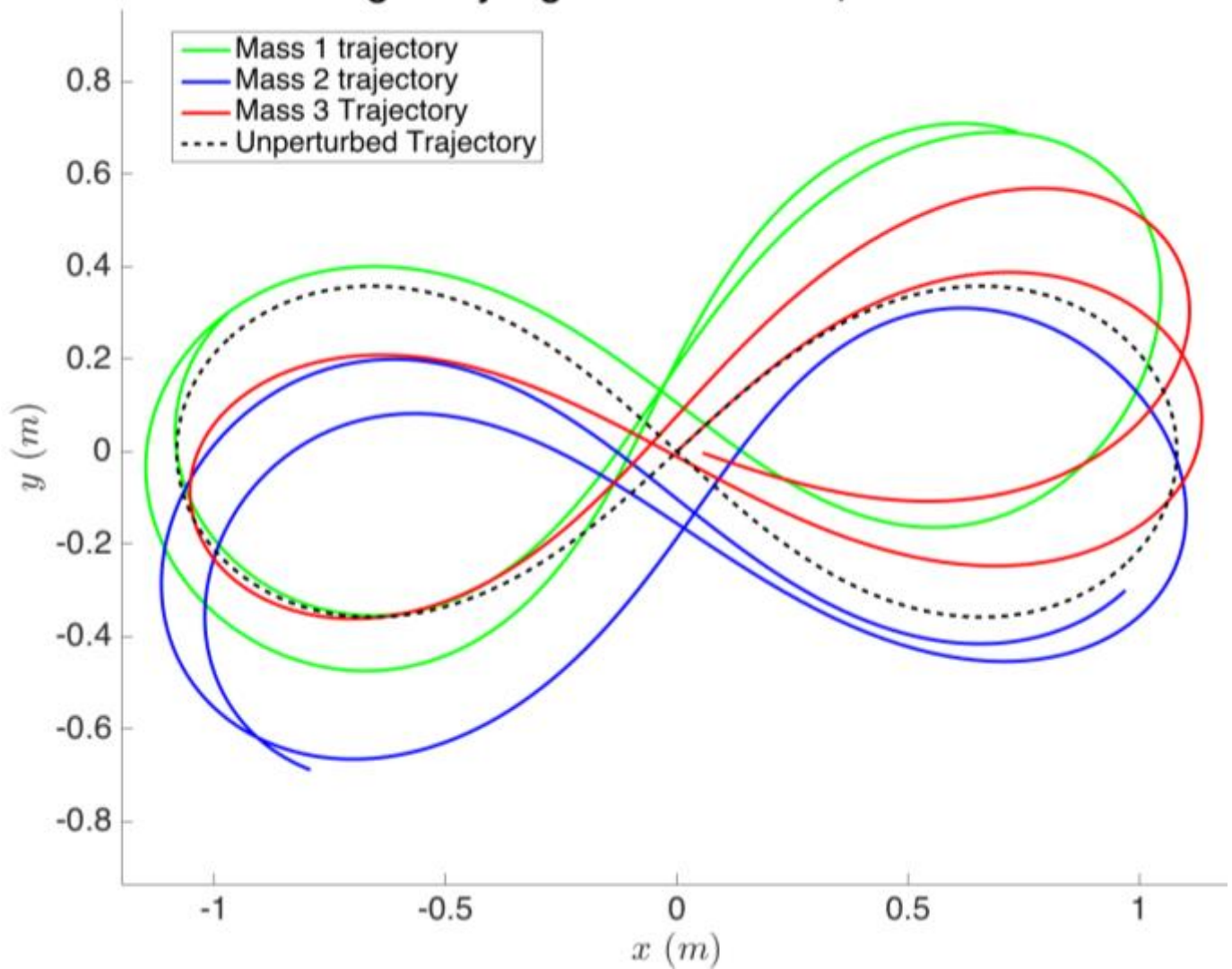
initsd = [r0_1; r0_2; r0_3; v0_1; v0_2; v0_3];
initsi = [initsd; inits_i];
opts.RelTol = 1e-6; opts.AbsTol = 1e-6;

% Integrate via ODE 45
[tarray, zarray] = ode45(@RHS_12d, tspan, initsd, opts, p);
x1 = zarray(:,1); y1 = zarray(:,2); x1i = zarray(:,13); y1i = zarray(:,14);
x2 = zarray(:,3); y2 = zarray(:,4); x2i = zarray(:,15); y2i = zarray(:,16);
x3 = zarray(:,5); y3 = zarray(:,6); x3i = zarray(:,17); y3i = zarray(:,18);

% Plotting
figure
hold on
axis equal
xlim([-1.2 1.2])
r1 = plot(x1, y1, 'g'); set(r1, {'LineWidth'}, {2});
r2 = plot(x2, y2, 'b'); set(r2, {'LineWidth'}, {2});
r3 = plot(x3, y3, 'r'); set(r3, {'LineWidth'}, {2});
% For comparison
r1i = plot(x1i, y1i, 'k--'); set(r1i, {'LineWidth'}, {2});
r2i = plot(x2i, y2i, 'k--'); set(r2i, {'LineWidth'}, {2});
r3i = plot(x3i, y3i, 'k--'); set(r3i, {'LineWidth'}, {2});
ylabel('$y$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
xlabel('$x$ (m)', 'Interpreter', 'latex', 'FontSize', 24);
set(gca, 'fontsize', 20)
title('Montgomery Eight: 10 time units, Perturbed IC', 'FontSize', 20);
legend('Mass 1 trajectory', 'Mass 2 trajectory', 'Mass 3 Trajectory', ...
      'Unperturbed Trajectory');

```

Montgomery Eight: 10 time units, Perturbed IC



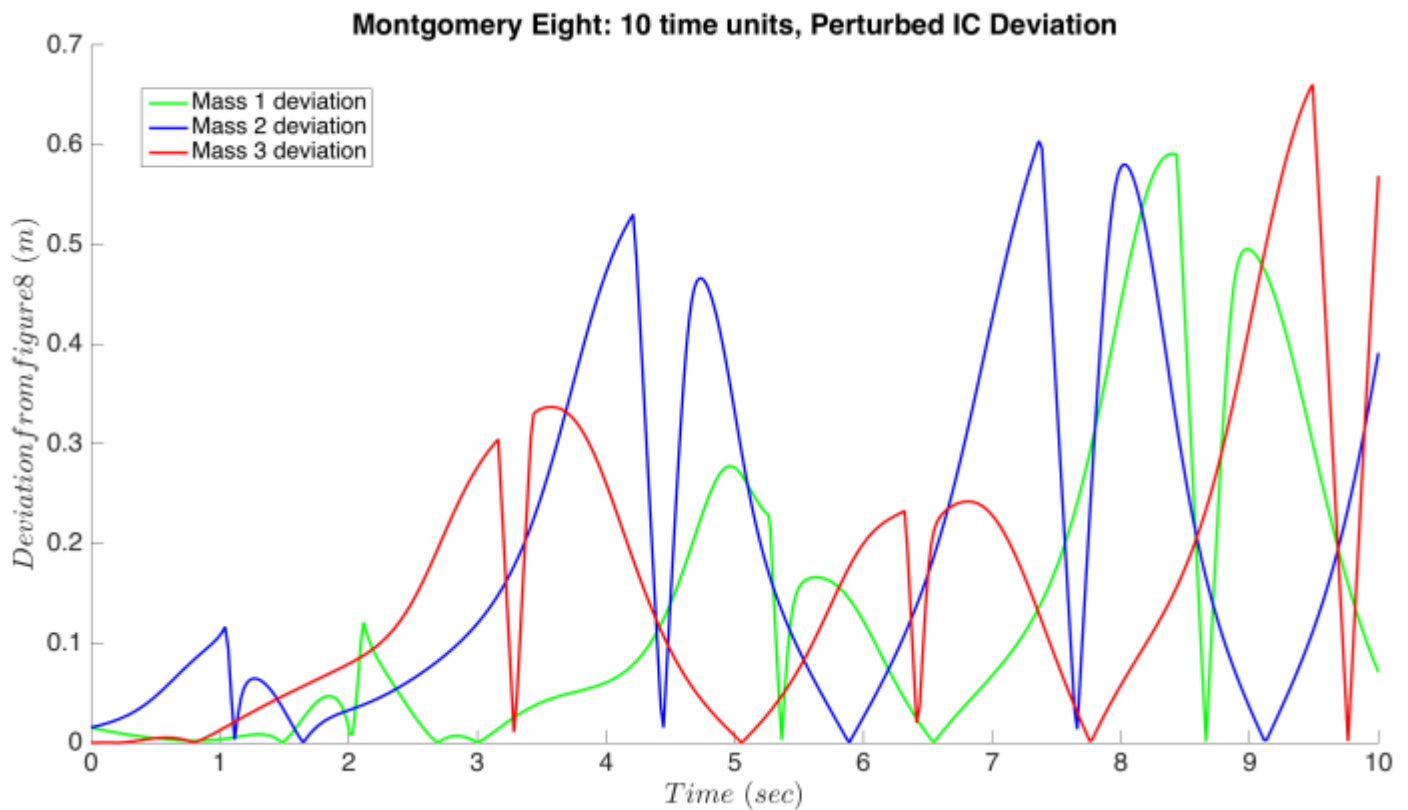
p12d: We see that a small perturbation (as evidenced in the initial conditions), causes the trajectories to deviate from the original figure 8 shape we found in parts a. and b. of this problem. By extending the time span to ten seconds, we see that the trajectory does not return to the original figure 8. We examine next if the deviation between the perturbed trajectory and original trajectory increases over time.

```

% Examine behavior of perturbed trajectory as t -> large
figure
hold on
r1pert = sqrt(x1.^2 + y1.^2);
r1i = sqrt(x1i.^2 + y1i.^2);
r2pert = sqrt(x2.^2 + y2.^2);
r2i = sqrt(x2i.^2 + y2i.^2);
r3pert = sqrt(x3.^2 + y3.^2);
r3i = sqrt(x3i.^2 + y3i.^2);

temp1 = plot(tarray,abs(r1pert-r1i),'g'); set(temp1,{ 'LineWidth'},{2});
temp2 = plot(tarray,abs(r2pert-r2i),'b'); set(temp2,{ 'LineWidth'},{2});
temp3 = plot(tarray,abs(r3pert-r3i),'r'); set(temp3,{ 'LineWidth'},{2});
ylabel('$Deviation from figure 8$ $(m)$','Interpreter','latex','FontSize',24);
xlabel('$Time$ $(sec)$','Interpreter','latex','FontSize',24);
set(gca,'fontsize',20)
title('Montgomery Eight: 10 time units, Perturbed IC Deviation','FontSize',20);
legend('Mass 1 deviation', 'Mass 2 deviation', 'Mass 3 deviation');

```



p12d: We see that the general trend of the deviation increases over time. This characterizes this system as a chaotic system (small change in initial conditions leads to drastic change in behavior).